

The economic burden of violence against women and violence against children in Italy

Gabriella Conti* Francesca Marazzi† Elena Pizzo‡ Emma Scandolo§
Flavia Bustreo¶

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Abstract

Violence against women (VAW) and violence against children (VAC) represent major public health and societal challenges, yet evidence on their economic burden in Italy remains limited. This study aims to quantify the health and productivity costs associated with VAW and VAC using nationally representative data and an econometric approach.

We combined survey and administrative data from multiple sources, including the Italian National Institute of Statistics (ISTAT), the European Working Conditions Survey (EWCS), hospital discharge records, and the Survey of Health, Ageing and Retirement in Europe (SHARE). We estimated the association between exposure to violence and a range of health and labour market outcomes using logistic and linear regression models, controlling for demographic characteristics. Estimated marginal effects were combined with unit cost data to derive outcome-specific costs. Lifetime costs per victim were computed by aggregating short-term and long-term effects. We conducted a probabilistic sensitivity analysis using Monte Carlo simulation to assess uncertainty.

Exposure to violence is associated with significant adverse health and economic outcomes. In the ISTAT sample, the estimated lifetime cost of VAW is €40,248 per victim, driven by both productivity losses (€19,424) and healthcare costs (€20,825). Workplace violence is associated with much higher lifetime costs, exceeding €478,000 per victim and mostly reflecting lost earnings, although this estimate rests on a small sample. Childhood violence is associated with higher rates of depression, smoking and limitations in daily activities in adulthood. Applying lifetime costs to the women who experienced violence in the previous 12 months, and adding the costs of hospital and emergency department admissions and of feminicides, the annual economic burden of VAW in Italy is estimated at €37.5 billion, equivalent to 1.71% of GDP, with a 95% uncertainty interval of €28.1 billion to €49.1 billion (1.28%–2.24% of GDP).

Violence against women and children is associated with a substantial and persistent economic burden on Italian society. The magnitude of these costs underscores the importance of prevention and intervention policies, particularly those addressing mental health and labour market consequences. Reducing violence could yield significant economic as well as social benefits.

*University College London, London, UK

†University of Rome Tor Vergata, Rome, IT

‡University College London, London, UK

§University College London, London, UK

¶Fondation Botnar, Basel, Switzerland

1 Introduction

Italy ranks high among EU countries for incidents of violence against women (VAW), with approximately 40% of women reporting at least one such episode. Regarding violence against children (VAC), about 27% of adults recall experiencing physical harm during childhood, and half report harm, neglect, or poor parental relationships. The COVID-19 pandemic only exacerbated this phenomenon: the average number of women accessing anti-violence services almost doubled from the 2016-18 average of 1306 women to 2496 on average in March and April 2020. Data collected by the Italian National Institute of Statistics (ISTAT) indicates a 59% increase in calls to a hotline for Gender-Based Violence (GBV) victims in March and April 2020 compared to the same period in 2019.

A few works in the literature have tried to assess the costs and consequences of GBV. For example, Sabia et al. (2013), using a US sample, find that women who experienced sexual violence have a 6.6% lower probability of labour force participation and 5.1% lower average wages. Adams-Prassl et al. (2024) focus on violence between colleagues in Finland and find a striking asymmetry: male perpetrators experience substantially weaker labour market consequences after attacking female colleagues than after attacking male colleagues. Furthermore, women experiencing forced sexual intercourse are found to develop depression with a higher likelihood and have reduced academic performance (Rees and Sabia, 2013). Other works have tried to assess the overall burden of GBV: Spence (2020) provides cost estimates for Rwanda (with costs for victims amounting to \$10,355,812.97 and government costs amounting to \$13,082,542.07).

Evidence on the economic costs of violence in Italy remains scarce. Existing estimates of the costs of VAW and VAC rely mainly on technical reports which are not based on econometric analyses and suffer from several data limitations (see Section 2). For example, Badalassi et al. (2013), utilising data from 2006, use an accounting approach with economic and social multipliers to estimate the cost of inaction at €16.72 billion.

With this study, we aim to fill this gap. Our main contribution is to ground the costing of violence in econometric evidence. Existing estimates for Italy attribute costs to victims on the basis of assumptions about the services they use and the consequences they suffer, or extrapolate them from other countries. We instead use nationally representative Italian survey data to estimate how exposure to violence is associated with a wide range of health and labour market outcomes, comparing victims with individuals who did not experience violence but have similar observed background characteristics, and distinguishing short-term from long-term associations where the data allow. These estimated differences in outcomes, rather than assumed service use or unit costs borrowed from another country, determine which costs we attribute to violence, and we then value them using Italian unit costs. Combining survey and administrative data, we obtain lifetime costs per victim and the total annual burden of violence, and we assess the uncertainty around our estimates through a probabilistic sensitivity analysis. To our knowledge, these are also the first estimates for Italy to cover both violence against women and violence against children within a common framework.

2 Background and previous Italian evidence

2.1 Violence against women and children in Italy

Violence against women and children is pervasive worldwide. Almost one in three women aged 15 and over, around 736 million women, have experienced physical or sexual violence by an intimate partner or sexual violence by a non-partner at least once in their lifetime (WHO, 2021). In the European Union, the 2024

EU gender-based violence survey found that 30.7% of women had experienced physical violence (including threats) and/or sexual violence in their lifetime; the corresponding figure for Italy is 31.7% (FRA, EIGE and Eurostat, 2024). The 2014 ISTAT survey on violence against women, which we use in this paper, shows that 31.5% of women aged 16–70 have experienced physical or sexual violence at least once since the age of 16 (Section 3). First results from the 2025 edition of the survey, which covers women aged 16–75, report a very similar share (31.9%) (ISTAT, 2025).

Evidence on violence against children is more fragmented, as Italy lacks a national data-collection system on child maltreatment (Tronci et al., 2022). The main source is the national survey of municipal social services carried out by Terre des Hommes and Cismai for the Italian Authority for Children and Adolescents. In its third edition, 374,310 children and adolescents were in the care of social services at the end of 2023 (42 per 1,000 resident minors), of whom 113,892 (30.4%) were in care because they were victims of maltreatment (Terre des Hommes and Cismai, 2025). Since they only capture cases known to social services, these figures are a lower bound of the extent of the phenomenon.

Since 2013, Italy has substantially strengthened its legal framework on violence against women. In 2013 Italy ratified the Council of Europe Convention on preventing and combating violence against women and domestic violence (Istanbul Convention), which also commits Parties to collect data on violence at regular intervals and to support research on its causes and effects (Council of Europe, 2011, Article 11). In the same year, Law No. 119/2013 introduced, among other measures, the urgent removal of offenders from the family home, financial support for a national network of anti-violence centres and shelters, and an extraordinary national action plan against sexual and gender-based violence. Subsequent reforms introduced paid leave for women victims of gender-based violence (Legislative Decree No. 80/2015), protections for children orphaned by domestic crimes, including femicide (Law No. 4/2018), and stronger protection of victims together with stricter penalties for domestic violence, stalking and other gender-based crimes (Law No. 69/2019, the so-called *Codice Rosso*). Law No. 168/2023 further strengthened the protection of women victims of violence and domestic violence, and Law No. 181/2025 introduced femicide as a separate offence in the Criminal Code (Article 577-bis). Victims can rely on the national 1522 helpline, active since 2006, and on a network of anti-violence centres and shelters run mainly by women’s organisations. Nonetheless, many prevention and protection measures are set out in policy documents rather than in legislation, the system lacks a coordinated multi-agency response, and there is little systematic monitoring of whether the national action plans have been implemented (Tronci et al., 2022).

2.2 Previous estimates of the costs of violence in Italy

Studies of the economic costs of violence typically distinguish between direct costs, i.e. the value of the goods and services used to respond to violence (such as healthcare, police, justice, social services and shelters); indirect costs, such as lost productivity; and intangible costs of pain and suffering. They also differ in their approach, which can be based on accounting, econometric or macroeconomic methods (Ashe et al., 2017). Only a handful of studies have applied these methods to Italy.

The first comprehensive estimate of the cost of VAW in Italy is the study by Badalassi et al. (2013), sponsored by the Deputy Minister of Labour and Social Policies with responsibility for Equal Opportunities. Using an accounting approach, the study combines data from the 2006 ISTAT survey on violence against women, covering around 25,000 women aged 16–70, with interviews with professionals in services dealing with violence and with women who had experienced violence. It estimates a total annual cost of €16.72 billion, of which €1.77 billion are direct costs (healthcare, law enforcement, justice, social services and anti-

violence centres), €0.6 billion are economic multiplier effects such as lost productivity, and €14.3 billion are non-monetary costs and social multiplier effects, valued by simulating the compensation for physical, moral and biological damage. By contrast, spending on prevention and awareness-raising amounted to only €6.3 million. Several cost components could not be calculated because of missing or inconsistent data, and the estimates are not based on econometric models of the consequences of violence. Ciaschini and Chelli (2021) use the direct costs estimated by Badalassi et al. (2013), including public spending on services for victims, in a macroeconomic input–output model in which these services are represented as an additional industry, and simulate the output gains and losses for the economy of reallocating this spending to other industries.

More recently, the European Institute for Gender Equality estimated the costs of gender-based violence in all EU Member States (EIGE, 2021). The estimates are built in two steps. First, the costs for the United Kingdom are obtained by multiplying the Home Office unit costs of each category of crime (Heeks et al., 2018), which cover lost economic output, health and criminal justice services, and the physical and emotional impact on victims, by the number of incidents estimated mainly from the Crime Survey for England and Wales; smaller costs for civil justice, social welfare and specialist services are added from expenditure data. Second, the UK total is extrapolated to each Member State in proportion to its population size. For Italy, this yields a cost of €38.9 billion for gender-based violence against women and €20.4 billion for intimate partner violence against women in 2019. In the underlying UK estimates for women, the largest cost component is the monetised physical and emotional impact on victims (55.6%), followed by the criminal justice system (20.4%) and lost economic output (13.9%). The cost attributed to each incident is therefore a UK parameter rather than an estimate of the consequences of violence for Italian victims, and the extrapolation by population size does not reflect the Italian prevalence of violence, Italian wages or the costs of Italian services.

For violence against children, the only national estimate we are aware of is a study carried out by Bocconi University for Terre des Hommes and CISMAI (Terre des Hommes and CISMAI, 2013). Using an accounting approach, it estimates that the public costs associated with the prevalence of child maltreatment amount to €13.1 billion a year, or 0.84% of GDP, while the costs generated by each year’s new cases of maltreatment amount to €910 million (0.06% of GDP); both shares refer to 2010 GDP.

Taken together, existing estimates for Italy share a common limitation: none of them estimates, on Italian microdata, how the outcomes of victims differ from those of comparable non-victims. Accounting approaches (Badalassi et al., 2013; Terre des Hommes and CISMAI, 2013) sum the costs of the services used by victims and of consequences valued through assumptions or parameters drawn from other sources; EIGE (2021) applies UK unit costs per incident; and Ciaschini and Chelli (2021) take earlier accounting estimates as given. These studies also treat violence against women and violence against children separately, and EIGE (2021) relies on data from another country. This paper addresses these limitations by providing econometric evidence for the costing analysis, applying the two-step approach that Conti et al. (2021) developed for child maltreatment in the UK to Italian data on both VAW and VAC (Section 4).

3 Data

For our study, we collect data from various sources, from surveys to administrative records, as well as reports by non-governmental organisations. Table 1 describes our data sources, including how violence is defined and their time and age coverage; the variables used from each source are listed in Table 6 in the Appendix.

Table 1: Data sources

Data source	Violence definition	Coverage	Respondents
ISTAT <i>Indagine multiscopo sulle famiglie</i> (IMF)	Self-reported experiences of violence	2014	Adults
ISTAT Violence Against Women Survey	Self-reported exposure to physical/sexual violence since age 16; timing of last episode	2014	Age: 16–70
European Working Conditions Survey (EWCS)	Self-reported experiences of workplace violence (sexual harassment, physical violence, bullying)	2010–2015	Workers aged 15+
Hospital discharge data	Diagnoses related to violence	2004–2016	All ages
EMUR (EMergency–URgency) data flow	Diagnoses related to violence	2024	All ages
<i>Casa delle donne</i> reports	Femicide cases collected from newspapers	2012–2022	All ages
Survey of Health, Ageing and Retirement in Europe (SHARE)	Self-reported physical abuse, neglect or poor relationship with parents before age 17	2017 (retrospective)	50+

Violence against women (VAW) To study violence against women, we draw on the 2014 ISTAT Violence Against Women Survey, a nationally representative dataset covering women aged 16–70.¹ The variables derived from ISTAT and used in our analysis are summarised in Table 6. In this survey, we define VAW as exposure to physical and/or sexual violence since age 16, and we exploit the timing of the last episode to distinguish between three categories: no violence, violence within the past 12 months, and violence more than 12 months before the interview. This distinction allows us to estimate separately the short-term and long-term associations between violence and the outcomes listed in Table 6 in the Appendix. Overall, 31.5% of respondents report having experienced VAW, and 4.5% report an episode within the previous 12 months. The ISTAT survey also includes the productivity and health outcomes listed in Table 6. On the productivity side, we use information on employment, part-time employment among employed women, and receipt of unemployment benefits. On the health side, we use indicators for insomnia, depression, headaches, nausea, heart conditions, chronic conditions, and weakness. Background characteristics—including age group, education level, relationship status, citizenship, and region of residence—are included as controls, in line with the measurement definitions in Table 6.

To analyse violence occurring in the workplace, we supplement our ISTAT data with the 2010–2015 European Working Conditions Survey (EWCS), which provides the additional measures listed in Table 6. In EWCS, respondents are explicitly asked whether they experienced sexual harassment, physical violence, or bullying *within the past 12 months*. Thus, workplace VAW in this dataset reflects only recent (i.e., short-term) experiences of violence. The prevalence of workplace VAW among female workers in Italy is 2.5%. Since EWCS does not collect information on the timing of episodes beyond the 12-month window, we can estimate only the short-term effects of workplace violence. To derive long-term cost estimates, we assume that the short-term effects remain constant over the longer term. EWCS further provides the continuous wage measure used to study productivity, as well as indicators for physical pain, headaches, and mental

¹Data available at: <https://www.istat.it/en/press-release/violence-against-women/>

health problems. Background characteristics include age, years of education, company size and sector, and parental citizenship.

We use the Italian hospital discharge data for the period 2004–2016 to estimate the costs of hospital admissions recording violence or abuse as a primary or secondary cause of admission. The selection of violence-related diagnoses is based on a screening of the ICD-9-CM codes (International Classification of Diseases, 9th Revision, Clinical Modification) associated with each hospitalisation. The set of codes associated with violence or abuse is the same one used in the technical report of ISTAT (2020) and is listed in Table 7 in the Appendix. Using the same set of codes, we reconstruct the number of violence-related Emergency Department (ED) admissions using the Italian EMUR (EMergency–URgency) data flow for 2024.²

With respect to the most extreme form of violence against women, i.e., feminicides, in the absence of an official time series covering the whole period, we first have to build a series of the number of cases that occurred to then estimate their societal cost. We resort to the reports issued yearly by the non-profit organisation ‘Casa delle donne per non subire violenza’, which collects data on cases of feminicides from Italian newspapers.³ For this paper, we have digitised the number of victims (and their age) for the period 2012–2022.

Violence against children (VAC) Data on violence against children come from the Survey of Health, Ageing and Retirement in Europe (SHARE). SHARE Wave 7 asks respondents aged 50+ to recall childhood experiences before age 17, including neglect, physical abuse, and indicators of poor parental relationships. Consistent with previous studies (Conti et al., 2021; Brugiavini et al., 2023), we combine these items into a global binary measure of VAC. To assess long-term consequences, we link the VAC measure to the adult outcomes listed in Table 6: depression (mental health), smoking (health behaviour), and reduced ability to perform daily activities (productivity losses). As with the other datasets, we include the corresponding background characteristics—age, citizenship, gender, parental background, and childhood socioeconomic conditions—as control variables. To complement this self-reported information, we also identify violence-related hospital and ED admissions of patients under 18 in the hospital discharge records and EMUR data.

4 Methods

To compute the costs of VAW and VAC on health and socioeconomic outcomes, we adopt the two-step procedure of Conti et al. (2021), which can be summarised as follows:

1. *Econometric analysis*: we use ordinary least squares (OLS) and logit models to estimate the association between VAW or VAC and health and socioeconomic outcomes, controlling for demographic characteristics.
2. *Cost analysis*: we combine the regression results from (1) with the best available published evidence on the unit cost of each outcome.

4.1 Econometric analysis

To estimate the association between violence and subsequent health and productivity outcomes, we use the variables summarised in Table 6. Because outcomes differ across surveys, our modelling approach depends

²Since, for privacy reasons, the number of admissions per municipality is not disclosed when positive but below three, we randomly assign a number of admissions between one and three to each municipality with missing data.

³Yearly reports are available at <https://femicidiocasadonne.wordpress.com/ricerche-pubblicazioni/>.

on their measurement. All binary outcomes (such as depression, insomnia, employment status, receipt of unemployment benefits, or physical pain) are analysed using a logistic regression model, while the only continuous outcome in our data (log wages) is analysed using a linear regression model. As explained in Section 3, where the data allow, we distinguish between short- and long-term exposure to violence. In the ISTAT survey, respondents report the timing of their most recent episode of violence, which allows us to estimate separately the short-term effects of violence (i.e. within one year of the episode) and the longer-term effects that may persist or emerge thereafter. In the EWCS, respondents are asked whether they experienced workplace violence within the 12 months prior to the survey. Hence, we estimate short-term associations for workplace violence and obtain long-term effects by assuming that these short-term effects remain constant over time. In SHARE, violence refers to adverse experiences during childhood (before age 17). Because this measure reflects exposure occurring many years before the interview, it directly captures long-term effects. Formally, for each binary outcome y_i we estimate the following logistic model:

$$\Pr(y_i = 1 \mid V_i, X_i) = \Lambda(\alpha + \beta_1 \mathbb{1}\{V_i = \text{short-term}\} + \beta_2 \mathbb{1}\{V_i = \text{long-term}\} + X_i' \gamma), \quad (1)$$

where V_i denotes exposure to violence, X_i is the vector of background characteristics, and $\Lambda(\cdot)$ is the logistic function. In datasets where only a binary violence measure is available, the model reduces to:

$$\Pr(y_i = 1 \mid V_i, X_i) = \Lambda(\alpha + \beta V_i + X_i' \gamma). \quad (2)$$

For the continuous wage outcome, we estimate the linear specification:

$$\log(w_i) = \alpha + \beta V_i + X_i' \gamma + \varepsilon_i. \quad (3)$$

To facilitate interpretation, all logistic regression results are presented as *average marginal effects*, which express the change in the probability of each outcome when moving from no exposure to violence to each violence category (or from 0 to 1 in the binary case). For the linear wage regression, we interpret the coefficient β directly as the percentage change in wages associated with violence.

4.2 Costing framework

We estimated the economic costs of violence against women (VAW) and violence against children (VAC) using a societal perspective, capturing both direct healthcare costs and indirect productivity losses. The costing analysis builds on the econometric results described above, combining estimated marginal effects with unit cost information drawn from the literature and administrative sources. For each outcome, the cost per victim was computed as the product of the estimated marginal effect of violence on the probability of that outcome and the corresponding unit cost. This approach yields the expected cost associated with violence for each outcome. Outcomes for which the estimated association is not statistically significant at the 10% level are not costed. The societal perspective includes both the costs for the Italian National Health Service (*Servizio Sanitario Nazionale*, NHS) and the costs for society, in terms of productivity losses and out-of-pocket costs.⁴ Where data allow, we distinguish between short-term and long-term costs. Short-term costs

⁴Costs are presented in 2024 prices and inflated and converted where necessary, using standard methods. When lifetime costs are assessed, the costs following the first year are discounted using a 3% discount rate. For the analysis, we use an average age at which the violence occurs (based on each database), and we extrapolate assuming constant effects over time. For productivity loss costs, we assume effects last until retirement age, while for health outcomes, we assume effects last until death (using the average life expectancy in Italy).

refer to outcomes occurring within the year following an episode of violence, while long-term costs capture persistent effects from the second year onwards until retirement or death, depending on the nature of the outcome. Total cost per victim was then obtained by summing across all relevant health and productivity components. Health-related costs include expenditures borne by the NHS, such as treatment for depression, insomnia, headaches, and other conditions associated with violence. Productivity losses were estimated based on reduced labour market participation, increased part-time work, unemployment benefits, and wage reductions. Unit costs for each outcome were obtained from published sources and national statistics, and expressed or inflated in 2024 euros where appropriate. For VAW using ISTAT data, we estimate both short-term and long-term effects separately based on the timing of the most recent episode of violence. Lifetime costs per victim are computed as the sum of short-term costs and the discounted stream of long-term costs. Productivity losses are calculated up to retirement age, while healthcare costs are accumulated over the lifetime.

For workplace violence (EWCS), only short-term effects are directly observed. To estimate lifetime costs, we assume that short-term effects persist over time and apply them over the relevant time horizon.

For VAC, exposure occurs before age 17 and all estimated effects reflect long-term consequences in adulthood; therefore, all costs are treated as lifetime costs.

Additional cost components We complement the regression-based estimates with additional cost components that are not directly captured by the econometric analysis. These include hospitalisation costs, ED admissions and the societal cost of feminicides.

Hospitalisation costs were estimated using Italian hospital discharge data, identifying admissions associated with violence through ICD-9-CM codes. The average cost per admission was derived using national tariffs linked to Diagnosis-Related Groups (DRGs).⁵

The cost of ED admissions was estimated using the same set of ICD-9-CM codes as for hospitalisations, using the EMUR data flow for the year 2024. ED unit costs were estimated using an Italian macro-area approach based on standard regional classification (North, Centre, South, Islands), reflecting known differences in healthcare expenditure, hospital production costs, and reimbursement structures. Unit costs were conservatively set at €450 for Northern regions, €400 for Central regions, €350 for Southern regions, and €360 for Islands.

The cost of feminicides was estimated as the discounted lifetime productivity loss per victim, calculated from the average age at death (50 years old) to the average life expectancy. We assumed full employment and applied average wages for the relevant age group.

Aggregation to population level Total annual costs were obtained by multiplying the estimated cost per victim by the number of affected individuals in the population. For VAW, the number of victims was obtained by applying the share of women who report an episode of violence in the 12 months before the interview (4.5%) to the population of women aged 16–70 in Italy. Hospitalisation, ED and feminicide costs were aggregated using the corresponding numbers of admissions and cases. For VAC, the number of victims was derived from prevalence estimates and population counts for victims aged under 17 in Italy. Costs were aggregated across all components and expressed both in absolute terms and as a percentage of national gross domestic product (GDP).

⁵The national tariffs associated with every DRG code are retrieved from the Italian Ministry of Health’s website at https://www.salute.gov.it/portale/temi/p2_6.jsp?lingua=italiano&id=3662&area=programmazioneSanitariaLea&menu=vuoto.

4.3 Sensitivity analysis

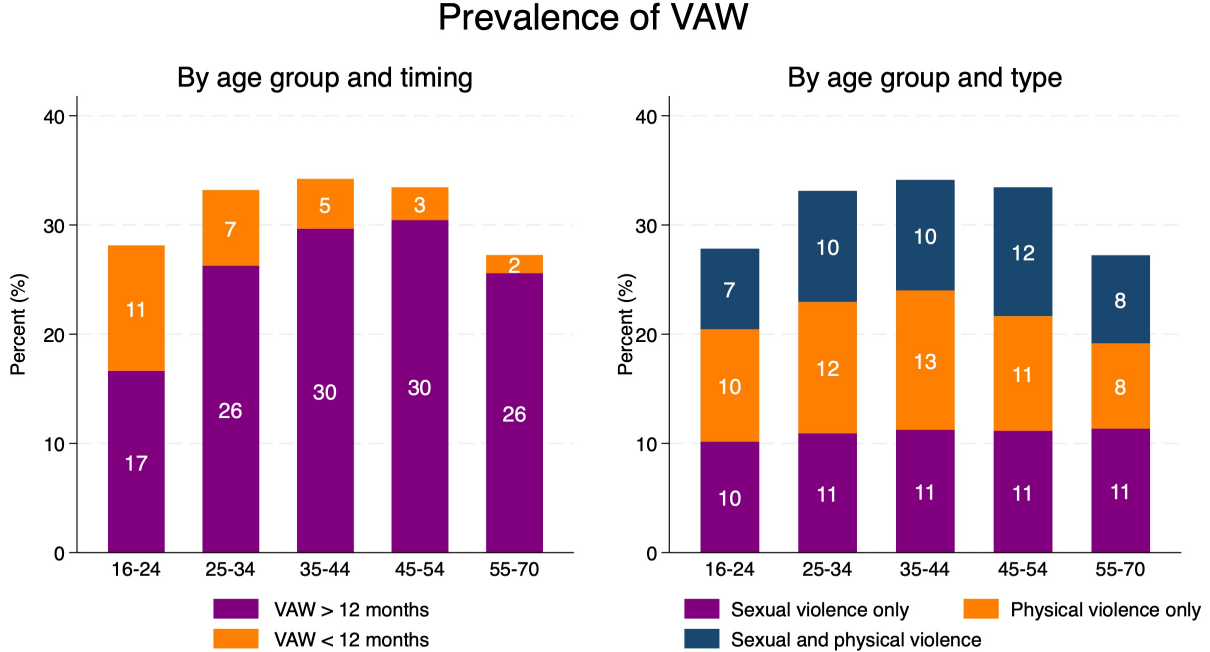
Probabilistic analysis We conducted a probabilistic sensitivity analysis (PSA) using Monte Carlo simulation (5,000 iterations). Regression coefficients were modelled as normally distributed using estimated coefficients and standard errors, while cost parameters were modelled using gamma distributions. For each iteration, total costs were recalculated and aggregated to the population level. We report mean estimates and 95% uncertainty intervals based on the simulated distribution. Cost inputs were modelled using lognormal distributions, reflecting the non-negative and right-skewed nature of cost data. For each cost component, the mean was set equal to the deterministic estimate reported in the main analysis, and uncertainty was parameterised using a coefficient of variation (CV) of 20%, consistent with standard practice in health economic evaluation. The parameters of the lognormal distributions were derived from these moments. Since the regression outputs had already been translated into monetary cost estimates per victim, uncertainty in econometric parameters (e.g. marginal effects) was implicitly incorporated within the variability assigned to the cost components. All cost components were assumed to be independently distributed, due to the absence of information on the covariance structure across estimates. While this assumption may underestimate joint uncertainty, it is commonly adopted when covariance matrices are unavailable. In each simulation draw, total cost per victim was computed as the sum of all cost components. Aggregate annual costs were then obtained by multiplying simulated per-victim costs by the estimated number of victims in the population. The number of femicide cases was treated as fixed, and associated costs were modelled using a lognormal distribution consistent with the per-case estimate. Total costs were finally expressed as a share of GDP, which was held constant at the 2024 estimate reported by ISTAT. The PSA results are summarised using the mean and the 2.5th and 97.5th percentiles of the simulated distribution, corresponding to a 95% uncertainty interval. This approach captures the combined effect of uncertainty in the cost inputs on the final cost estimates; the numbers of victims and of femicide cases, and GDP, are held fixed. PSA results are reported for VAW.

5 Results

5.1 Prevalence and consequences of VAW

Prevalence Before turning to the regression results, we first document the prevalence of VAW in our data. Figure 1 shows the prevalence of VAW in the ISTAT survey by age group, distinguishing both the timing and type of violence. Overall, nearly one in three women reports having experienced physical or sexual violence at least once since the age of 16. The age gradient in prevalence reflects clear life-cycle patterns. Among young women aged 16–24, 11% report an episode occurring within the 12 months prior to the interview, and a further 17% report an episode occurring more than one year earlier. As age increases, the share of recent victims declines steadily, whereas the prevalence of long-term episodes rises, resulting in the highest cumulative prevalence among women aged 35–44 and 45–54. The right-hand panel of Figure 1 further distinguishes between sexual violence only, physical violence only, and the combination of both. Sexual violence displays a relatively stable prevalence across age groups, with roughly 10–11% of women affected. Physical violence is more concentrated among women in their mid-thirties to mid-forties, where 12–13% report having experienced physical violence only. Combined sexual and physical violence affects between 7% and 12% of women, with the highest prevalence again in the 35–44 and 45–54 groups.

Turning to workplace violence, Figure 2 shows descriptive evidence based on the EWCS. Although overall



Source: ISTAT, 2014

Figure 1: Prevalence of VAW

prevalence is lower than for VAW in the ISTAT survey—2.5% among female workers in Italy—the age pattern is stark. Young workers aged 16–24 are disproportionately affected: 2% report experiencing only physical violence, 2% report bullying only, and 2% report exposure to more than one type of violence in the workplace. Prevalence declines sharply with age, and by age 35 and above, almost all reported cases involve bullying only. The concentration of workplace violence among the youngest workers suggests that early labour market entrants face a particularly high risk of harassment and intimidation in the workplace.

Consequences Table 2 summarises the estimated associations between violence and the health and productivity outcomes described in Section 3, together with the corresponding monetary costs per victim. All reported coefficients are average marginal effects from logistic regressions, except for the wage regression in the EWCS, where the coefficient represents the percentage change in wages associated with exposure to violence.

Panel A reports the short-term associations, defined as outcomes in the year following an episode of VAW. We find sizeable and statistically significant associations with both productivity and health. Women who experienced violence within the past 12 months are 4.2 percentage points more likely to work part-time, conditional on being employed, reflecting a short-term reduction in productivity. The effects on unemployment benefits are not statistically significant. Short-term health consequences are substantial. Recent victims are 18.3 percentage points more likely to report insomnia, 17.5 percentage points more likely to suffer from depression, and 10.8 percentage points more likely to report headaches. These increases translate into non-negligible economic costs per victim, ranging from approximately €315 for headaches to €1,479 for productivity losses associated with depression. We also test for, but do not find, statistically significant effects on the probability of being employed and on the following health conditions: nausea, heart

Table 2: VAW: regression estimates and costs for victims

	Regression estimate	Cost per victim
Panel A: Short-term costs of VAW in the year after violence		
	ISTAT	
Productivity loss from increased PT work when employed	0.042**	€926.88
Productivity loss from increased unemployment income	NSS	NSS
Cost of insomnia	0.183***	€481.69
Cost of depression (NHS)	0.175***	€549.51
Cost of depression (prod loss)	0.175***	€1,478.90
Cost of headache	0.108***	€314.71
Panel B: Costs of VAW from year 2 after violence onwards		
	ISTAT	
Productivity loss from increased PT work when employed ^a	NSS	NSS
Productivity loss from increased unemployment income ^a	0.003*	€65.04
Cost of insomnia ^b	0.106***	€7,637.86
Cost of depression (NHS) ^b	0.078***	€6,704.84
Cost of depression (prod loss) ^b	0.078***	€18,044.69
Cost of headache ^b	0.08***	€6,381.66
Panel C: Short-term costs of violence in the workplace		
	EWCS	
Cost of reduced wage in the year after	-0.254*	€18,955.51
Cost of physical pain	0.211**	€516.95
Cost of headache	0.172**	€501.21
Cost of mental health	0.363***	€1,857.47
Panel D: Short-term costs of hospital and ED admissions		
	Hospital discharge data and EMUR	
Cost of hospital admission via DRG code	NA	€2,219
Cost of ED admission ^c	NA	€415.75

Regression estimates in Panels A–C are average marginal effects of switching from 0 to 1 in exposure to violence on the probability of various outcomes considered, besides the only continuous outcome (log wages) where the estimate reports the percentage change in wages associated with violence. All ISTAT regressions control for age (5 categories), citizenship, education level (4 categories), geographic area (5 categories) and current and past relationship status (7 categories). All EWCS regressions control for age (continuous), citizenship, education years, and company characteristics (private or public, size). PT: part-time. NSS: not statistically significant. NA: not applicable. Sample size: 24,761 women (ISTAT); 763 (EWCS wages) - 1410 (other EWCS outcomes). Panel D computes the average cost of hospital admissions recording violence or abuse as a primary or secondary cause of admission, using NHS national tariffs and hospital admission DRG codes, and the average cost of violence-related ED admissions, using region-specific unit costs by macro-area. * p<0.10, ** p<0.05, *** p<0.001

^a calculated up to retirement age

^b lifetime cost until death

^c based on regional data of ED admissions for violence (in adults) in 2024

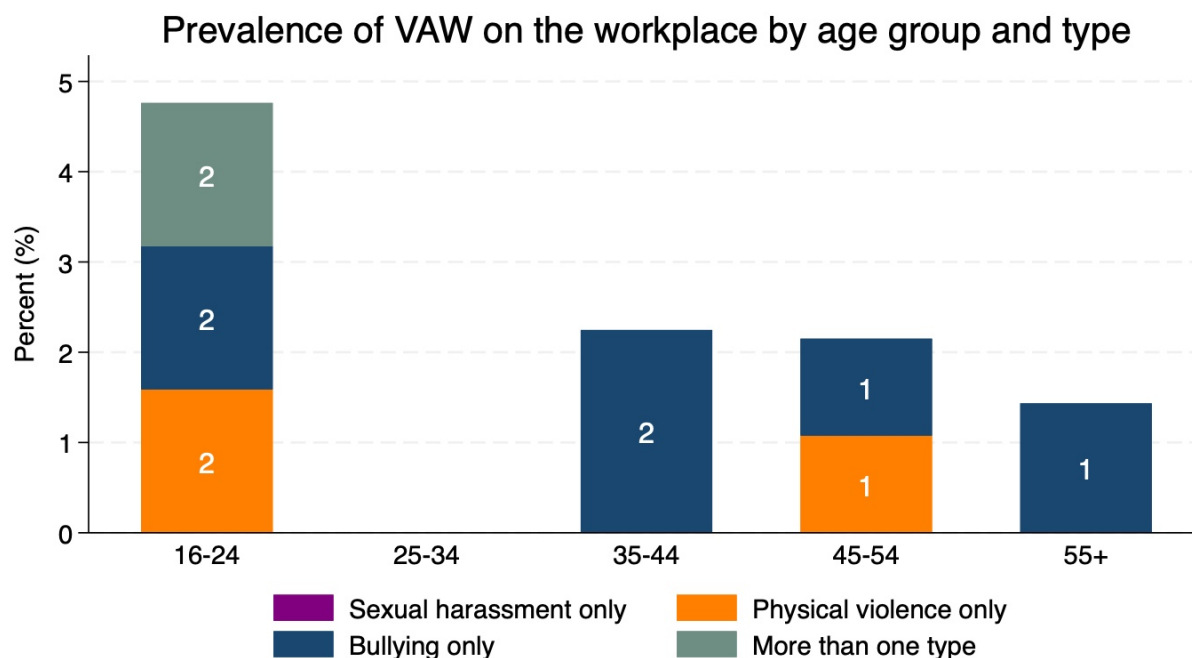


Figure 2: Prevalence of VAW in the workplace

conditions, chronic conditions, and weakness.

Panel B presents the estimated effects from the second year after violence onwards. The short-term effect on part-time work attenuates over time and becomes statistically insignificant, whereas long-term victims are 0.3 percentage points more likely to receive unemployment benefits. Long-term health effects remain sizeable. Women continue to experience significantly higher rates of insomnia (10.6 percentage points), depression (7.8 percentage points), and headaches (8 percentage points) many years after the last episode of violence. Because these are lifetime consequences, the associated costs are correspondingly higher. For example, lifetime productivity losses linked to depression exceed €18,000 per victim, and long-term healthcare costs (NHS) also remain substantial.

Panel C reports the effects of violence occurring in the workplace. Since EWCS records only episodes experienced in the 12 months prior to the survey, these coefficients capture short-term impacts. Exposure to workplace violence is associated with a 25.4% reduction in wages in the following year, a substantial productivity loss with an implied annual cost of nearly €19,000 per victim. Workplace violence is also associated with a higher probability of physical pain (21.1 percentage points), headaches (17.2 percentage points), and mental health problems (36.3 percentage points), imposing a substantial economic burden.

Lastly, Panel D reports the average cost to the NHS of hospital and ED admissions. Hospital admissions are costed using the national tariffs associated with the DRG code of each admission (see Section 4.2). Assessing the total cost of hospital admissions over the period covered by the discharge data, we estimate an average cost for the National Health Service of €2,219 per admission (inflated to 2024). ED admissions recorded in the 2024 EMUR data are instead costed by applying region-specific unit costs, set by macro-area (North, Centre, South and Islands), to the number of violence-related ED admissions in each region. This

yields an average cost of €416 per ED admission.

5.2 Lifetime costs of VAW

Table 3 reports the lifetime monetary losses associated with VAW, combining both productivity losses and healthcare-related costs. For the ISTAT sample, lifetime costs are derived by combining the short-term costs in the year immediately after the episode of violence with the discounted costs estimated from year 2 onwards. For workplace violence in the EWCS, lifetime costs are computed differently due to data limitations. Since EWCS only records whether violence occurred within the 12 months preceding the interview, we can estimate only short-term effects. To extrapolate lifetime costs, we assume that the effect in the first year after violence persists at the same magnitude from year 2 onwards.

Table 3: VAW: lifetime costs per victim

	Regression estimate	Cost per victim
Panel A: Lifetime costs of VAW^c		
	ISTAT	
Lifetime productivity loss from increased PT work when employed ^a	0.042;0	€926.88
Lifetime productivity loss from increased unemployment income ^a	0;0.003	€57.59
Lifetime cost of insomnia ^b	0.183;0.106	€7,660.52
Lifetime cost of depression (NHS) ^b	0.175;0.078	€6,851.40
Lifetime productivity loss from depression ^b	0.175;0.078	€18,439.11
Lifetime cost of headache ^b	0.108;0.080	€6,312.83
Total lifetime productivity loss		€19,423.58
Total lifetime health-related cost		€20,824.75
Total loss		€40,248.33
Panel B: Lifetime costs of violence in the workplace^d		
	EWCS	
Lifetime productivity loss from reduced wage ^a	-0.254*	€406,732.13
Lifetime cost of physical pain ^b	0.211**	€12,914.39
Lifetime cost of headache ^b	0.172**	€12,521.13
Lifetime cost of mental health ^b	0.363***	€46,403.16
Total lifetime productivity loss		€406,732.13
Total lifetime health-related cost		€71,838.68
Total loss		€478,570.81
Panel C: Lifetime costs of feminicides		
	Casa delle donne	
Lifetime productivity loss	NA	€1,350,384.96

Estimates are average marginal effects of switching from 0 to 1 in exposure to violence on the probability of various outcomes considered, besides the only continuous outcome (log wages) where the estimate reports the percentage change in wages associated with violence. All ISTAT regressions control for age (5 categories), citizenship, education level (4 categories), geographic area (5 categories), and current and past relationship status (7 categories). All EWCS regressions control for age (continuous), citizenship, education years, and company characteristics (private or public, size). PT: part-time. NA: not applicable. Sample size: 24,761 women (ISTAT); 763 (EWCS wages) - 1410 (other EWCS outcomes). * p<0.10, ** p<0.05, *** p<0.001

^a calculated up to retirement age

^b lifetime cost until death

^c two regression estimates are reported for ISTAT lifetime costs: the first one refers to the year after violence (Panel A of Table 2), and the second one refers to the estimate from year 2 after violence (Panel B of Table 2); estimates that are not statistically significant are set to 0

^d EWCS lifetime costs are calculated using the short-term costs for the year after violence, and discounted short-term costs until retirement or death.

Our estimates show that the total lifetime economic burden of violence is considerable. The total lifetime loss per victim using ISTAT data is €40,248 (Table 3, Panel A). The costs associated with workplace violence,

i.e., estimated using EWCS, are approximately €480,000 per victim (Panel B), of which about €407,000 are lost earnings.

Finally, we estimate the costs associated with feminicides. As described in Section 4.2, we assume that all victims were employed and estimate that the average age of the victims recorded in the ‘Casa delle donne per non subire violenza’ is 50 years old. The average discounted lifetime productivity loss due to femicide is €1.3 million per victim.

5.3 Prevalence and consequences of VAC

Figure 3 describes the prevalence of VAC in SHARE. Just over half of respondents (55% of men and 53% of women) report at least one form of childhood adversity: physical abuse, neglect, or a poor relationship with their parents. A noteworthy share of respondents report experiencing more than one type of adversity (27–28%), although men more often report physical abuse only (13% vs 7%) and women neglect only (17% vs 12%).

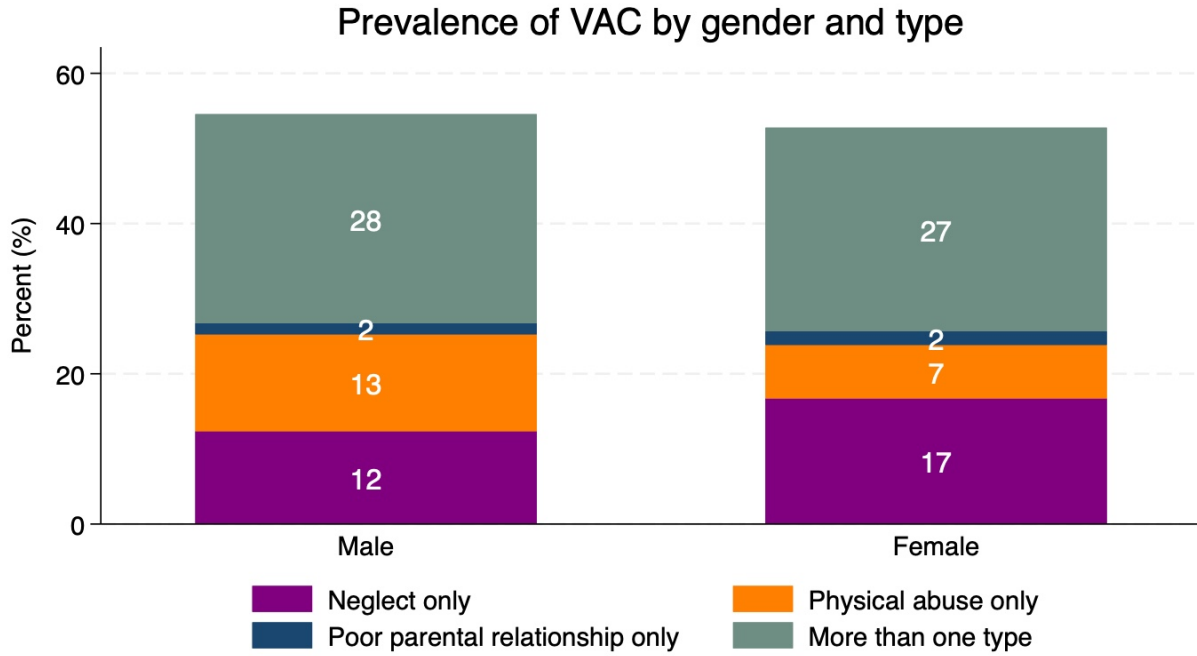


Figure 3: Prevalence of VAC

Panel A of Table 4 presents the estimated *lifetime* costs associated with VAC, based on the SHARE data. Unlike VAW, where short- and long-term effects can be distinguished, VAC represents adversity experienced before age 17 and therefore its consequences unfold entirely over the life course. For this reason, all costs reported in Panel A of Table 4 are lifetime estimates. The regression results indicate that exposure to childhood violence is associated with a significantly higher likelihood of depression, smoking, and reduced ability to carry out daily activities in adulthood. These outcomes translate into substantial long-term economic costs. For depression, the combined lifetime burden—including NHS costs, productivity losses, and out-of-pocket expenses—amounts to nearly €6,000 per affected individual. Smoking, which is 6.6 percentage

Table 4: VAC: regression estimates and costs per victim

	Regression estimate	Cost per victim
Panel A: Lifetime costs of VAC in adulthood		
	SHARE data	
Discounted cost of depression	0.024**	
Lifetime NHS costs ^a		€2,877.61
Lifetime productivity losses ^b		€2,749.82
Lifetime out-of-pocket costs ^a		€345.85
Total lifetime cost of depression		€5,973.29
Discounted cost of smoking	0.066**	
Lifetime direct and indirect costs of smoking ^c		€4,539.53
Discounted cost of reduced productivity	0.02**	
Lifetime productivity loss from reduced daily activities ^d		€49,591.97
Lifetime disability benefits ^a		€3,133.96
Panel B: Short-term costs of hospital and ED admissions		
	Hospital discharge data and EMUR	
Cost of hospital admission via DRG code	NA	€1,694
Cost of ED admission ^e	NA	€428.50

Estimates are average marginal effects of switching from 0 to 1 in exposure to violence on the probability of various outcomes considered. All SHARE regressions in Panel A control for age, citizenship, gender, parental background, and childhood socioeconomic conditions. NA: not applicable. Sample size: 1806-2043 respondents aged 50+. Panel B computes the average cost of hospital admissions recording violence or abuse as a primary or secondary cause of admission, using NHS national tariffs and hospital admission DRG codes, and the average cost of violence-related ED admissions, using region-specific unit costs by macro-area. * p<0.10, ** p<0.05, *** p<0.001

^a from 18 years old to death

^b from average starting working age until retirement (25–67)

^c from 14 years old to death

^d from average starting working age until death

^e from data on regional ED admissions for violence (in minors) in 2024

points more likely among VAC victims, generates lifetime direct and indirect costs of around €4,540. The largest component arises from reduced productivity: VAC increases the probability of limitations in daily activities by 2 percentage points, corresponding to a lifetime productivity loss of nearly €50,000, in addition to disability-related costs exceeding €3,000.

Panel B reports that the average cost per hospital admission associated with VAC is €1,694 (inflated to 2024), and the average cost per ED admission associated with VAC is €428.50, using the same approach as for VAW. These are short-term costs due to the nature of the data that we use.

5.4 Total VAW and VAC burden and share of Italian GDP

We aggregate the component-level estimates described in the previous sections to obtain separate measures of the total annual economic burden of VAW and of VAC in Italy, expressed both in absolute terms and as a share of GDP (Table 5).

The primary costs of VAW are drawn from the ISTAT survey. As documented in Table 3, the estimated lifetime cost per victim amounts to €19,423.58 in productivity losses and €20,824.75 in healthcare costs. To translate these per-victim figures into an annual population-level cost, we apply the estimated prevalence of recent victimisation, defined as an episode occurring within the 12 months preceding the interview, to the relevant reference population. In the ISTAT survey, 4.5% of respondents report such a recent episode. Applying this prevalence measure to the 2024 population of women aged 16–70 in Italy (20,613,385), we obtain an estimate of approximately 927,602 women who experienced violence in the past 12 months. Multiplying the per-victim lifetime costs by this annual victim count yields total productivity losses of €18,017,351,655 (0.82% of GDP), and total healthcare costs of €19,317,075,991 (0.9% of GDP). Because some of these women may also have experienced violence in previous years, applying lifetime costs to all of them may overstate the annual cost of new victimisation.

We complement the regression-based estimates of VAW with hospitalisation costs derived from the Italian hospital discharge records, using DRG-linked national NHS tariffs. As reported in Table 2, the average cost per violence-related admission is €2,219.32 (inflated to 2024 euros), and the average annual number of admissions involving women is 228, yielding an annual hospitalisation cost of €506,004.96. The average cost per violence-related ED admission is €416 (in 2024 euros), and the number of ED admissions involving women in 2024 is 21,243, yielding an annual cost of €8,831,860.

We separately estimate the societal cost of feminicides. Drawing on annual reports from the NGO ‘Casa delle donne’, which collects data on feminicide cases from Italian newspapers, we record 104 feminicide cases in 2024. As described in Section 4.2, the cost per case is estimated as the discounted lifetime productivity loss from the average age of the victim (50 years) to life expectancy, assuming full employment and applying average age-specific wages. This yields a cost per victim of €1,350,384.96 and an aggregate annual cost of feminicides of €140,440,035.81, equivalent to 0.01% of GDP. In total, we estimate a VAW cost of €37,484,208,558.62, equivalent to 1.71% of the Italian GDP in 2024.

We also aggregate VAC costs to the population level. The lifetime costs of depression, smoking, reduced productivity and disability have been estimated as the product of the cost per victim and the number of children who experienced violence in 2024. These costs represent 0.10%, 0.08%, 0.85% and 0.05% of GDP, respectively. We complement the regression-based estimates of VAC with hospitalisation costs derived from the Italian hospital discharge records, using DRG-linked national NHS tariffs. As reported in Table 4, the average cost per violence-related admission is €1,694 (inflated to 2024 euros), and the average annual number of admissions involving children is 416, yielding an annual hospitalisation cost of €704,843. The

average cost per violence-related ED admission is €428.50 (in 2024 euros), and the number of ED admissions involving children in 2024 is 3,766, yielding an annual cost of €1,613,740. In total, we estimate a VAC cost of €23,673,214,429.33, equivalent to 1.08% of the Italian GDP in 2024.

We exclude workplace violence costs estimated using EWCS data because there is substantial overlap with outcomes used in the ISTAT survey. Since working women are a subset of the broader female population already covered by ISTAT, aggregating both would amount to counting the same types of cost twice for an overlapping group of women. The EWCS estimates are therefore best interpreted as providing an alternative, complementary perspective on the intensity of costs for the specific subpopulation of working women exposed to workplace violence, rather than as an additional cost component to be summed. Since the EWCS measure also includes bullying, which the ISTAT measure of physical and sexual violence does not capture, this exclusion is conservative.

Table 5: Total costs as a share of GDP

	Cost per victim ^a	Total Cost ^{b,c}	As % of GDP ^d
Panel A: VAW			
Lifetime productivity loss	€19,423.58	€18,017,351,655	0.82%
Lifetime health-related cost	€20,824.75	€19,317,075,991	0.9%
Hospitalisation costs	€2,219.32	€506,004.08	0.0%
ED admission costs	€415.75	€8,831,860.00	0.00%
Femicide costs	€1,350,384.96	€140,440,035.81	0.01%
Total cost VAW		€37,484,208,558.62	1.71%
Panel B: VAC			
Lifetime cost of depression	€5,973.29	€2,235,860,870.31	0.10%
Lifetime cost of smoking	€4,539.53	€1,669,192,264.37	0.08%
Lifetime cost of reduced productivity	€49,591.97	€18,562,771,939.93	0.85%
Lifetime cost of disability	€3,133.96	€1,173,070,772.03	0.05%
Hospitalisation costs	€1,694.33	€704,842.68	0.00%
ED admission costs	€428.50	€1,613,740	0.00%
Total cost VAC		€23,673,214,429.33	1.08%

^a Lifetime productivity and healthcare losses are taken from Table 3 (ISTAT); for hospitalisation and ED admissions, cost per admission.

^b The number of VAW victims is estimated as 4.5% (share of women reporting violence in the past 12 months) of 20,613,386 women aged 16–70 in 2024, i.e. 927,602 women. Hospitalisation and ED costs use the numbers of admissions reported in the text. The total number of victims for VAC are estimated as (prevalence) out of 374,310 children in 2024.

^c Total number of femicides in Italy in 2024 is 104, as per ‘*Casa delle donne*’ report.

^d Italian GDP in 2024 was €2,192,182,000,000.

5.5 Results of the sensitivity analysis

We conducted a PSA using Monte Carlo simulation (5,000 iterations) to assess the sensitivity of our VAW cost estimates to parameter uncertainty. The PSA indicates that the short-term cost of VAW is €3,747 per victim (95% uncertainty interval: €3,053–€4,596). Uncertainty remains moderate, with mental health-related costs representing the main source of variability. The results indicate that the total annual economic burden of VAW in Italy is €37.5 billion, with a 95% uncertainty interval ranging from €28.1 billion to

€49.1 billion. Expressed as a share of national output, this corresponds to 1.71% of GDP (95% uncertainty interval: 1.28%–2.24%). The mean estimate closely matches the deterministic baseline. While uncertainty is non-negligible, the economic burden remains substantial across all simulations, with even the lower bound indicating a substantial macroeconomic impact.

Figure 4 shows the distribution of simulated lifetime costs per victim of VAW. The distribution is mildly right-skewed and centred around €40,000, with most simulated values between about €30,000 and €55,000.

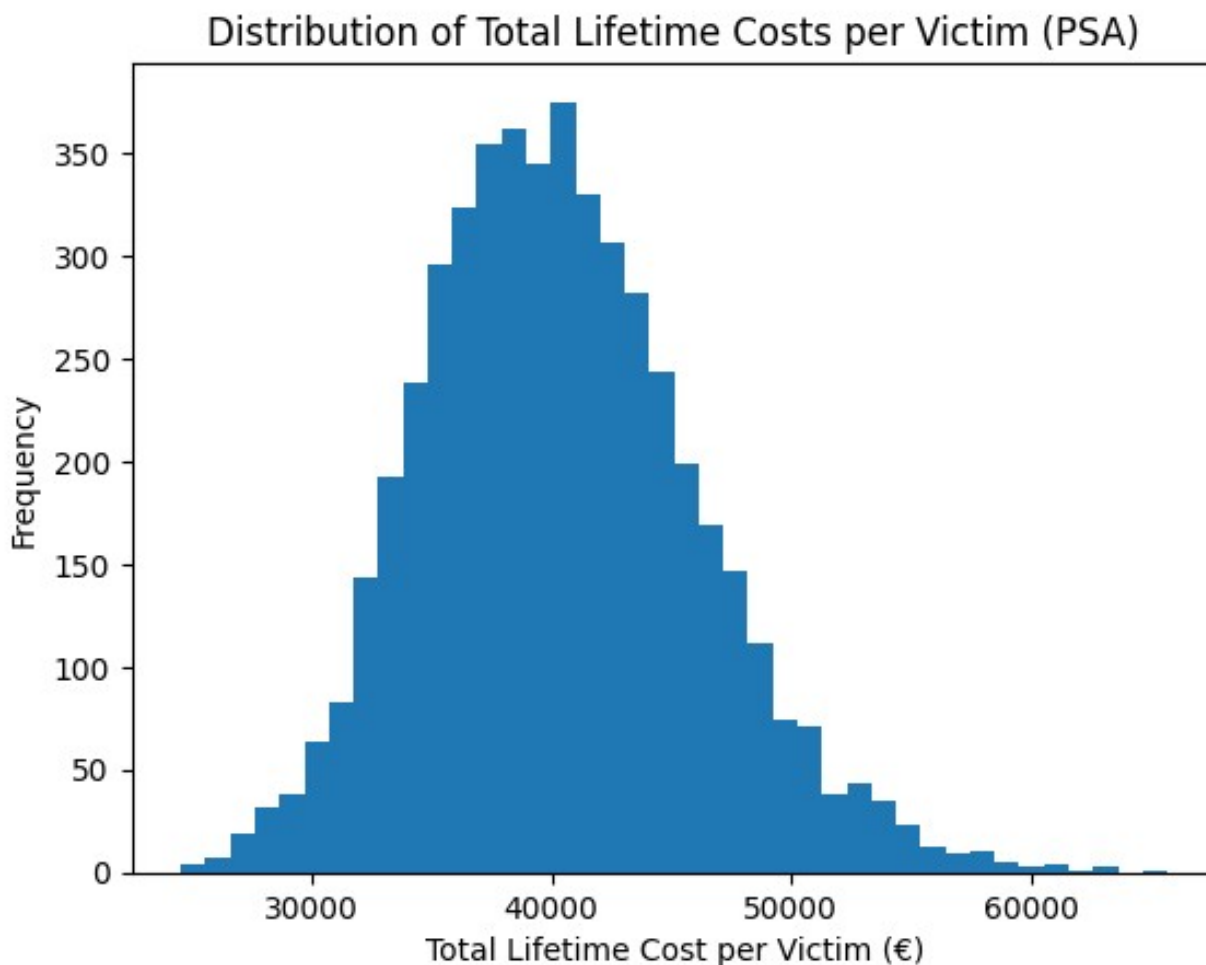


Figure 4: Sensitivity analysis cost distribution

6 Discussion

To our knowledge, this paper provides the first econometrically grounded estimates of the economic burden of violence against women and violence against children in Italy. Combining nationally representative survey data with administrative records, we find that exposure to violence is associated with large and persistent deteriorations in health, particularly mental health, and with smaller and shorter-lived differences in labour market outcomes. These translate into substantial costs for victims, the NHS and society as a whole. We estimate the lifetime cost of VAW at around €40,000 per victim, split almost equally between productivity losses and health-related costs. Applied to the approximately 930,000 women who experienced violence in the previous 12 months, and adding the costs of hospital and ED admissions and of feminicides, this implies an annual burden of €37.5 billion, or 1.71% of Italian GDP in 2024. The PSA indicates that, under the parameter uncertainty modelled, this burden remains substantial: even the lower bound of the 95% uncertainty interval, €28.1 billion, corresponds to 1.28% of GDP.

Three features of these results stand out. First, mental health accounts for most of the estimated costs. Depression and insomnia together account for more than 80% of the lifetime cost per victim of VAW, and depression alone accounts for lifetime productivity losses of more than €18,000 per victim. These associations attenuate but persist. While the association with part-time work is no longer statistically significant after the first year, women whose last episode of violence occurred more than a year before the interview remain significantly more likely to suffer from insomnia, depression and headaches. As a result, and under our assumption that long-term effects persist until retirement or death, most of the cost of VAW accrues in the years after the violence has ended rather than in its immediate aftermath: short-term costs account for less than a tenth of the lifetime total (about €3,750 out of €40,248 per victim).

Second, violence in the workplace is associated with particularly large costs for the women who experience it. Together with higher rates of physical pain, headaches and mental health problems, the estimated wage reduction implies a lifetime cost of nearly €480,000 per victim, of which about €407,000 are lost earnings, more than ten times the per-victim cost estimated from the ISTAT survey. This figure should be interpreted with caution: it rests on the assumption that the wage reduction persists until retirement and the health effects until death, and the wage estimate is based on a relatively small sample of 763 women and is significant only at the 10% level. As explained in Section 5.4, we do not add these costs to the aggregate burden, since working women are already covered by the ISTAT survey. Nonetheless, these estimates point to potentially large labour market consequences for women exposed to violence at work, which deserve further research with larger samples.

Third, the costs of violence against children extend well into adulthood. Respondents aged 50 and over who report adverse experiences before age 17 (physical abuse, neglect or a poor relationship with their parents) are more likely to suffer from depression, to smoke, and to be limited in their daily activities, decades after the exposure. By contrast, violence-related hospital and ED admissions of children cost less than €3 million a year. Hospital and emergency records alone would therefore capture only a small fraction of the economic consequences of VAC, most of which emerge decades later.

Our estimate for VAW is considerably larger than the previous Italian estimate, the €16.72 billion cost of inaction calculated by Badalassi et al. (2013) using 2006 data, and close to the €38.9 billion (in 2019) that EIGE (2021) obtains for Italy by extrapolating costs estimated for the United Kingdom. Neither figure is directly comparable with ours. More than four fifths of the total in Badalassi et al. (2013) consists of non-monetary costs and social multiplier effects, while the EIGE (2021) estimate is dominated by the monetised physical and emotional impact on victims and by criminal justice costs, neither of which we include, and does

not reflect Italian prevalence or prices. Our estimates instead rest on the differences in health and labour market outcomes between victims and non-victims that we estimate on Italian data, valued using Italian unit costs, expressed in 2024 prices and accumulated over the lifetime of victims. Although they measure different things, all three estimates put the annual cost of VAW to Italian society above €15 billion.

Finally, our estimates omit several cost components, including the costs borne by the criminal justice system, anti-violence centres and shelters, and social services, as well as intangible costs, all of which would add to the total. Other assumptions, discussed in Section 7, could bias the estimates in either direction.

7 Strengths and limitations

This study has several strengths in the estimation of the economic costs of violence. First, the costing framework is grounded in an econometric analysis that links exposure to violence with a wide range of health and productivity outcomes using nationally representative data. By combining regression-based marginal effects with unit cost information, we move beyond descriptive or accounting approaches and provide estimates that reflect the expected costs attributable to violence. Second, the adoption of a societal perspective allows us to capture both healthcare expenditures and productivity losses, providing a comprehensive assessment of the economic burden. Third, the analysis distinguishes between short-term and long-term effects where data allow, enabling the estimation of lifetime costs per victim. Finally, uncertainty in cost estimates is explicitly addressed through probabilistic sensitivity analysis, which indicates that the burden remains substantial under the parameter uncertainty modelled.

The study also has several limitations. First, our estimates are based on cross-sectional associations conditional on observed characteristics. They may partly reflect unobserved factors, reverse causality (for example, poor mental health increasing vulnerability to violence) or recall bias in retrospective reports of childhood adversity, and should not be interpreted as causal effects. Second, cost estimates rely on the availability and quality of unit cost data, which are drawn from multiple sources and may not fully reflect heterogeneity in healthcare utilisation or labour market outcomes. Third, the approach assumes that cost components are additive and independent, as information on the covariance structure across outcomes is not available. This may lead to either overestimation (if outcomes overlap) or underestimation (if costs are jointly determined) of total costs. Fourth, for workplace violence, only short-term effects are directly observed; lifetime costs are derived under the assumption that these effects persist over time, which may overstate or understate true long-term impacts. Fifth, some cost components, such as informal care, intangible costs (e.g. pain, suffering, and reduced quality of life), and intergenerational effects, are not included, which would make our estimates understate the true societal burden. Finally, the aggregation to population-level costs relies on prevalence estimates and simplifying assumptions that may not capture temporal dynamics or regional variation; in particular, applying lifetime costs to all women who report violence in the previous 12 months may overstate the annual cost of new victimisation.

Taken together, these limitations should be considered when interpreting the results. However, given that violence is consistently associated with poorer mental health across all three surveys, and that the VAW burden remains substantial across the simulations of the PSA, we believe that the overall conclusion of a large economic burden is unlikely to be overturned.

8 Policy implications

The findings of this study have important implications for public policy. First, the substantial economic burden associated with violence (over 1.7% of GDP for violence against women alone) highlights that violence is not only a social and public health issue, but also a major economic concern. Investments in prevention and early intervention should therefore be viewed not only as an ethical imperative, but also as potentially good value for money. Even modest reductions in prevalence could generate large savings for both the healthcare system and the wider economy.

Second, the results point to mental health as a key channel through which violence generates costs. Depression, insomnia, and other psychological conditions account for a large share of both healthcare expenditures and productivity losses. This suggests that strengthening access to mental health services for victims, including early diagnosis, timely treatment, and long-term support, could substantially mitigate the economic consequences of violence. In the Italian context, this could involve expanding the availability and integration of psychological support services within the NHS, as well as strengthening the role of anti-violence centres (*centri antiviolenza*) in providing coordinated care pathways.

Third, the potentially large productivity losses associated with workplace violence, although imprecisely estimated, point to the need for stronger labour market and organisational policies. Employers and institutions play a crucial role in prevention and response, including the implementation of anti-harassment policies, reporting mechanisms, and support systems for victims. In Italy, recent legislative frameworks, such as the introduction of paid leave for victims of gender-based violence (Legislative Decree No. 80/2015), provide an important foundation, but could be further strengthened through improved enforcement, employer awareness, and broader access across sectors. Policies that facilitate continued labour market participation, such as flexible work arrangements, job protection, and targeted employment support, may help reduce the long-term economic impact.

Fourth, the persistence of feminicides in Italy underscores the urgency of strengthening prevention and protection mechanisms. The human and economic cost of each feminicide highlights the importance of early identification of high-risk situations and timely intervention. This includes improving coordination between law enforcement, judicial systems, and support services, as well as strengthening risk assessment tools and protective measures for victims. Enhanced monitoring systems and better data integration could also support more effective policy responses.

Fifth, the long-term nature of these costs underscores the importance of early-life interventions. The substantial lifetime burden associated with violence against children suggests that policies aimed at preventing childhood adversity and supporting affected individuals can yield long-term economic returns. This includes strengthening child protection systems, parenting support programmes, and school-based interventions. In Italy, this could build on existing initiatives coordinated by social services and organisations such as *Terre des Hommes* and *CISMAI*, by improving coverage, early detection, and inter-institutional coordination.

Finally, the magnitude of the estimated costs supports the integration of violence prevention into broader economic and public health strategies. Cost-of-violence estimates such as those presented here can inform resource allocation, justify investment in prevention programmes, and support cross-sectoral coordination between health, social services, education, and labour market institutions. In the Italian setting, this aligns with the objectives of the National Strategic Plan on Male Violence against Women (*Piano strategico nazionale sulla violenza maschile contro le donne*), and highlights the importance of sustained funding and monitoring to ensure effective implementation. Future policies should prioritise both prevention and mitigation, recognising that reducing violence is likely to generate substantial social and economic benefits over the long

term.

9 Conclusions

Violence against women and children is not only a violation of human rights and a major public health problem, but also a substantial economic burden for Italian society. Using nationally representative survey data, administrative records and an econometric costing framework, this paper estimates that violence against women is associated with costs of around €37.5 billion a year in Italy, or 1.71% of GDP, and finds that violence against children is associated with substantial lifetime costs in adulthood. Most of these costs reflect long-lasting differences in mental health and productivity, which accrue many years after the violence has occurred.

These findings carry two broader implications. First, because so much of the cost of violence materialises over the long run, prevention and early support for victims are likely to generate economic returns in addition to their primary social and ethical value. Second, the lack of systematic data on violence in Italy, illustrated by the lack of a long official series on feminicides, limits the ability to monitor the problem and evaluate policy responses. Regular nationally representative surveys that record the timing of violence, together with linked administrative data on healthcare, employment and the justice system, would allow future studies to track how the burden of violence evolves and to assess which interventions are most effective in reducing it.

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Appendix

Additional tables

Table 6: Measures used

Variable	Variable type	Source
VAW		
VAW	Categorical (none, short-term, long-term)	ISTAT
VAW	Binary	EWCS
<i>Productivity</i>		
Employment	Binary	ISTAT
Part-time employment (among employed)	Binary	ISTAT
Unemployment benefits	Binary	ISTAT
Wage (log)	Continuous	EWCS
<i>Health</i>		
Insomnia	Binary	ISTAT
Depression	Binary	ISTAT
Headache	Binary	ISTAT
Nausea	Binary	ISTAT
Heart conditions	Binary	ISTAT
Chronic conditions	Binary	ISTAT
Weakness	Binary	ISTAT
Physical pain	Binary	EWCS
Headache	Binary	EWCS
Mental Health problems	Binary	EWCS
<i>Background characteristics</i>		
Age	Categorical (5 categories)	ISTAT
Age	Continuous	EWCS
Region of residence	Categorical (NUTS-II)	ISTAT
Italian citizenship	Binary	ISTAT
Italian citizenship	Binary	EWCS
Education level	Categorical (4 categories)	ISTAT
Education years	Continuous	EWCS
Relationship status (current and past)	Categorical (7 categories)	ISTAT
Employed in a public company	Binary	EWCS
Employed in a company with > 10 employees	Binary	EWCS
VAC		
VAC	Binary	SHARE
<i>Productivity</i>		
Reduced daily activities	Binary	SHARE
<i>Health</i>		
Depression	Binary	SHARE
Smoking	Binary	SHARE
<i>Background characteristics</i>		
Italian citizenship	Binary	SHARE
Gender	Binary	SHARE
Age	Continuous	SHARE
Parental Background (Citizenship, age, employment)	Binary	SHARE
Financial hardships in childhood	Binary	SHARE
Number of people per room in childhood	Binary	SHARE
Religious upbringing	Binary	SHARE

Table 7: List of ICD-9-CM codes used for the definition of cases of violence against women and children

ICD-9-CM codes	
<i>Other injuries and conditions due to external causes</i>	
9955	Child maltreatment syndrome
99550	Child abuse, unspecified
99551	Child emotional/psychological abuse
99552	Child neglect (nutritional)
99553	Child sexual abuse
99554	Child physical abuse
99555	Shaken baby syndrome
99559	Other child abuse and neglect
99580	Adult maltreatment, unspecified
99581	Adult physical abuse
99582	Adult emotional/psychological abuse
99583	Adult sexual abuse
99584	Adult neglect (nutritional)
99585	Other adult abuse and neglect
<i>Other personal history presenting hazards to health</i>	
V1541	History of physical abuse
V1542	History of emotional abuse
<i>Other family and psychosocial circumstances</i>	
V6111	Counseling for victim of spousal and partner abuse
V6112	Counseling for perpetrator of spousal and partner abuse
V6121	Counseling for victim of child abuse
V6122	Counseling for perpetrator of parent-child abuse
<i>Observation and evaluation for suspected conditions not found</i>	
V715	Observation following alleged rape or seduction
V716	Observation following other inflicted injury
V7181	Observation and evaluation for suspected abuse and neglect
<i>Homicide and injury purposely inflicted by other persons</i>	
E960	Homicide and injury purposely inflicted by other persons in; fight, brawl, rape
E9600	Unarmed fight or brawl
E9601	Rape
E961	Assault by corrosive or caustic substance, except poisoning
E962	Assault by poisoning
E9620	Assault by drugs and medicinal substances
E9621	Assault by other solid and liquid substances
E9622	Assault by other gases and vapors
E9629	Assault by unspecified poisoning
E963	Assault by hanging and strangulation
E964	Assault by submersion [drowning]
E965	Assault by firearms and explosives
E9650	Assault by handgun
E9651	Assault by shotgun
E9652	Assault by hunting rifle
E9653	Assault by military firearms
E9654	Assault by other and unspecified firearm
E9655	Assault by antipersonnel bomb
E9656	Assault by gasoline bomb
E9657	Assault by letter bomb
E9658	Assault by other specified explosive
E9659	Assault by unspecified explosive
E966	Assault by cutting and piercing instrument
E967	Perpetrator of child and adult abuse
E9670	Perpetrator of child and adult abuse, by father, stepfather, or boyfriend
E9671	Perpetrator of child and adult abuse, by other specified person
E9672	Perpetrator of child and adult abuse, by mother, stepmother, or girlfriend
E9673	Perpetrator of child and adult abuse, by spouse or partner
E9674	Perpetrator of child and adult abuse, by child
E9675	Perpetrator of child and adult abuse, by sibling
E9676	Perpetrator of child and adult abuse, by grandparent
E9677	Perpetrator of child and adult abuse, by other relative
E9678	Perpetrator of child and adult abuse, by non-related caregiver
E9679	Perpetrator of child and adult abuse, by unspecified person
E968	Assault by other and unspecified means
E9680	Assault by fire
E9681	Assault by pushing from a high place
E9682	Assault by striking by blunt or thrown object
E9683	Assault by hot liquid
E9684	Assault by criminal neglect
E9685	Assault by transport vehicle
E9686	Assault by air gun
E9687	Assault by human bite
E9688	Assault by other specified means
E9689	Assault by unspecified means
E969	Late effects of injury purposely inflicted by other person